

Multi-Seasonality Demand Forecasting for High-Volume For-Hire Vehicles in New York City

Manuel D’Alterio Grazioli 2104370 — Rebecca Esegio 2144564

June 2026

1 Introduction

1.1 Problem Definition and Objective

Demand forecasting is a key task in transportation systems, where resources must be allocated efficiently in response to continuously changing mobility patterns. Accurate forecasts support operational planning, improve service quality, and help providers anticipate periods of high or low demand.

This project focuses on forecasting the hourly demand for High-Volume For-Hire Vehicle (FHVHV) services in New York City. The objective is to analyze the temporal behavior of taxi demand and evaluate the effectiveness of different time series forecasting methods in predicting future demand levels.

1.2 Research Workflow

Following the initial "Problem Definition" phase, the project was structured into the following stages:

1. Data gathering and preparation;
2. Exploratory analysis;
3. Estimation, interpretation, and comparison of candidate time series models.

2 Data Gathering and Preparation

2.1 The Dataset

The analysis is based on the New York City High-Volume For-Hire Vehicle (FHVHV) trip records. The original dataset is distributed as a collection of monthly files that cover a one-year period. Each file contains individual ride-level observations, including pickup and drop-off timestamps, trip distance, travel duration, fare components, and ride characteristics.

Due to the large volume of data, each monthly file was processed separately. A preliminary cleaning procedure was applied to remove observations that were

not meaningful for demand forecasting purposes. Specifically, trips with non-positive distances, travel times shorter than 60 seconds, negative fares, and missing pickup timestamps were excluded.

After the cleaning stage, trips were aggregated at the hourly level within each monthly file. This aggregation step transformed the original trip-level records into a more compact hourly dataset suitable for time series analysis.

Finally, the twelve monthly aggregated datasets were merged into a single chronological dataset, covering the entire observation period.

Therefore, the resulting dataset consists of hourly observations, where the number of trips represents the target variable of the forecasting problem. Additional aggregated measures, such as average trip distance, average trip duration, average fares, tips, tolls, and other ride-related characteristics, were retained as candidate explanatory variables for the subsequent modeling stage.

3 Exploratory Analysis

3.1 Distribution of Demand

The exploratory analysis aims to examine the main characteristics of the hourly taxi demand data in order to support model selection, by understanding its distribution, identifying recurring temporal patterns, and detecting possible anomalies.

The summary statistics reveal substantial variability in demand during the observation period. Hourly demand ranges from a minimum of 85 trips to a maximum of 75272 trips, with an average of approximately 28234 trips per hour and a median of 30401 trips. The large difference between the minimum and maximum values suggests that demand may exhibit considerable variability over time.

3.2 Temporal Evolution of Demand

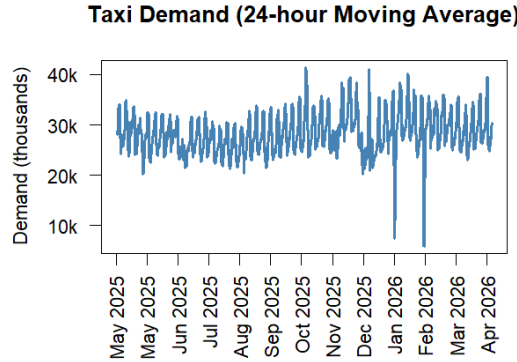


Figure 1: Temporal Evolution of Taxi Demand (24-hour Moving Average)

Figure 1 shows the evolution of taxi demand over the observation period, through a 24-hour moving average. The series fluctuates around a relatively stable level, generally ranging between 25000 and 35000 trips per day. Demand appears to increase slightly from the second half of the year 2025, reaching higher levels between October and December. Although the overall demand level remains relatively stable throughout the period, some pronounced temporary declines can be observed around January and February 2026. These episodes suggest periods of unusually low demand and motivate the subsequent outlier analysis. Overall, the plot suggests the presence of temporal variability while preserving a relatively consistent demand level throughout the period.

3.3 Outlier Analysis

To identify potentially anomalous observations, a formal outlier detection procedure based on standardized z-scores was applied. Specifically, observations with an absolute value greater than three standard deviations from the mean demand were flagged as potential outliers. This approach allows for the identification of unusually high or low demand values, relative to the overall distribution of the series.

The analysis identified four extreme observations, all corresponding to unusually high demand levels. These observations occurred during periods associated with major calendar events, including the daylight saving time transition and the New Year’s Eve and New Year’s Day celebrations. Consequently, the detected outliers are interpreted as actual demand shocks rather than measurement errors or data quality issues.

No observations with unusually low demand levels were identified according to the adopted criterion. This suggests that the temporary declines visible in the smoothed series before (Figure 1) are not extreme enough to be classified as statistical outliers in the original hourly data.

The detected outliers are retained in the dataset, as removing them would result in the loss of meaningful structural information about the system. Their inclusion preserves the real dynamics of the series, particularly in relation to extreme but informative demand fluctuations.

3.4 Daily Demand

To better understand the intraday dynamics of demand, the average number of trips was calculated for each hour of the day. The resulting profile, shown in Figure 2, highlights a clear daily cycle. Demand is at its lowest during nighttime hours, with a minimum amount reached at 5 AM. It gradually increases in the morning, stabilizing at higher levels from approximately 10 AM onward, reaching its peak in the evening around 8 PM. This pattern is consistent with typical urban mobility dynamics driven by commuting and leisure demand.

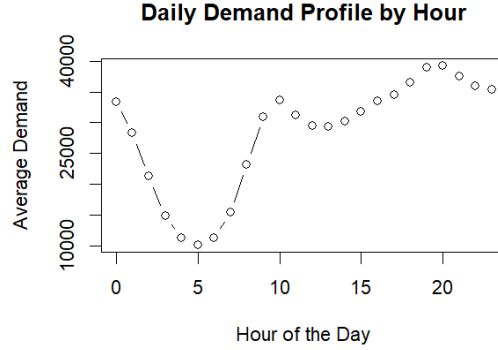


Figure 2: Daily Demand By Hour

3.5 Weekly Demand

Weekly demand patterns were also investigated using boxplots grouped by day of the week. Figure 3 shows that demand levels are relatively stable throughout the week.

Nevertheless, some differences emerge. In particular, Friday and Saturday exhibit slightly higher median demand levels and greater variability compared to the other days of the week. This pattern is consistent with increased leisure and social activities during weekends, which may generate additional demand for transportation services. Sunday also shows relatively high demand, although with a distribution more comparable to the rest of the week.

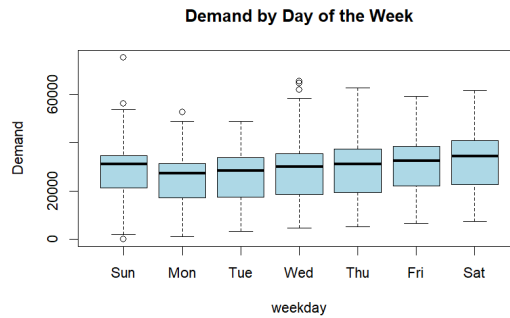


Figure 3: Weekly Taxi Demand

3.6 Exogenous Variables Diagnostics

Before building the forecast models, the candidate exogenous variables were examined to assess the degree of association among them. This preliminary

analysis is necessary to detect potential multicollinearity issues, which can lead to unstable parameter estimates.

To evaluate this aspect, both the correlation matrix and the Variance Inflation Factor (VIF) were computed. The correlation matrix (Figure 4) provides an initial overview of the dependency structure among the candidate explanatory variables, as well as their relationship with the target variable.

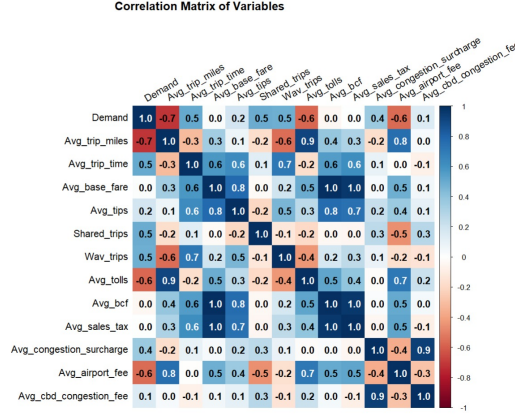


Figure 4: Correlation matrix

Overall, there are moderate to relatively strong correlations among several fare-related variables, which is expected since they capture similar aspects of trip pricing and structure. In particular, average base fare, tips, and trip time tend to move together, reflecting the common influence of trip length and intensity on different cost components. In the same way, toll and surcharge variables also show noticeable correlations, suggesting that these charges tend to vary together over time. Looking at the relationship with Demand, correlations are more mixed but still broadly consistent with economic intuition. Variables related to trip volume, such as shared and wav trips, are positively associated with demand, while fare-related components generally show weaker and less direct relationships. While some moderate correlations are present, there are no particularly high pairwise relationships among the regressors, suggesting that severe multicollinearity is not a concern at this stage. It is important to note that the correlation matrix is computed on the full set of available candidate variables in order to obtain an overall picture of their dependency structure. However, not all of these variables are included in following stages.

For the multicollinearity check, the Variance Inflation Factor (VIF) is computed on the subset of regressors that are effectively used in the forecasting models. This subset is selected a priori based on its relevance for demand dynamics and modelling consistency, excluding variables such as Avg_trip_miles, Avg_bcf, Avg_sales_tax, and Avg_cbd.congestion_fee, which are either redundant or not directly informative within the proposed framework.

The VIF results indicate that all selected regressors exhibit acceptable levels

of multicollinearity, with all values well below the commonly used threshold of 10. The highest values are observed for some fare-related variables, such as base fare and tolls, which is expected given their shared economic structure. However, these correlations remain moderate and do not affect the stability of the estimation.

Overall, the results confirm that the selected set of regressors can be safely used in the subsequent modeling phase.

4 Time Series Model Selection and Comparison

Following the exploratory analysis, the hourly demand observations were converted into a time series object with a seasonal frequency of 24, corresponding to the dominant daily cycle observed in the data. This representation enables the application of time series techniques, including autocorrelation analysis, and stationarity testing.

4.1 Autocorrelation Analysis

The temporal dependence structure is analyzed through the autocorrelation function (ACF) and partial autocorrelation function (PACF), reported in Figure 5.

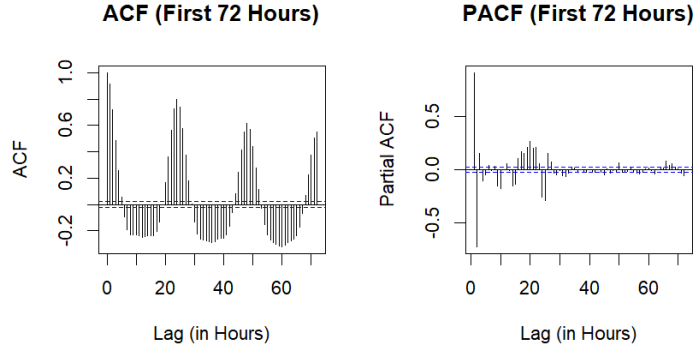


Figure 5: ACF and PACF of the hourly taxi demand series (first 72 Hours).

The ACF plot shows clear sinusoidal behavior with significant peaks at periodic intervals, suggesting oscillatory patterns within the series. The repeated peaks highlight the presence of a strong daily cycle, with peaks occurring approximately every 24 hours, reflecting the persistence of recurring travel demand patterns throughout the day.

The PACF plot shows dominant spikes at the first and second lags, followed by several smaller spikes and a gradual attenuation. This suggests that most of the relationships in the time series can be attributed to short-term dependencies, whereas seasonal dependence is particularly evident in the ACF.

4.2 Stationarity and Autocorrelation Tests

Stationarity is assessed using the Augmented Dickey-Fuller test. The results lead to the rejection of the null hypothesis, suggesting that the series is stationary from a statistical perspective. Finally, the Ljung-Box test is applied to assess whether the series behaves as white noise. The null hypothesis is strongly rejected, confirming the presence of significant autocorrelation in the data. This result further supports the existence of meaningful temporal structure. To evaluate the predictive performance of our forecasting models, we divided the dataset into training and testing subsets.

We allocated the initial 80% of the observations to the training set, while the test set consists of the final 20%, preserving the chronological approach of the temporal structure of the data.

It is important to specify that for all models incorporating exogenous regressors (such as SARMAX, Fourier-ARIMAX, and GAM), the evaluation on the test set was conducted as an ex-post evaluation. Instead of forcing the models to predict the future values of economic variables, such as Avg_fares and Avg_tips, we provided them with the actual data recorded during the test period. We made this choice to isolate the true performance of the models' mathematical structures. If we had forecasted the exogenous variables first, any error in those predictions would have affected the demand forecast, causing the compound error. By removing this uncertainty, we can fairly evaluate and compare how well each algorithm independently maps the relationship between external factors and taxi demand.

4.3 ARIMA and SARIMA

As a preliminary comparison, two non-seasonal ARIMA models were estimated. The model specification was guided by the patterns observed in the ACF and PACF plots reported in Figure 5.

The ACF displays an oscillatory pattern, while the PACF exhibits two dominant spikes at the first lags, followed by substantially smaller values. This behavior suggests the presence of short-term autoregressive dynamics, with the PACF indicating that low-order AR components may be relevant.

Since the Augmented Dickey-Fuller test rejected the null hypothesis of a unit root, the series was treated as stationary and no non-seasonal differencing was applied ($d = 0$).

Based on these initial indications, ARIMA(2,0,0) and ARIMA(2,0,1) models were estimated. The first specification captures the autoregressive structure suggested by the PACF, while the second introduces an additional moving-average component to account for potential short-term shock propagation. AIC values are nearly identical for the two specifications (132502.4 and 132219.9 respectively).

Diagnostic checks revealed that both models have residuals that do not resemble white noise. In particular, the residual ACF still exhibits several significant autocorrelations, with not substantial differences between the two model specifications (Figure 6) and, as we hypothesized, the Ljung-Box test strongly rejects the null hypothesis of white-noise residuals (p -value < 0.05).

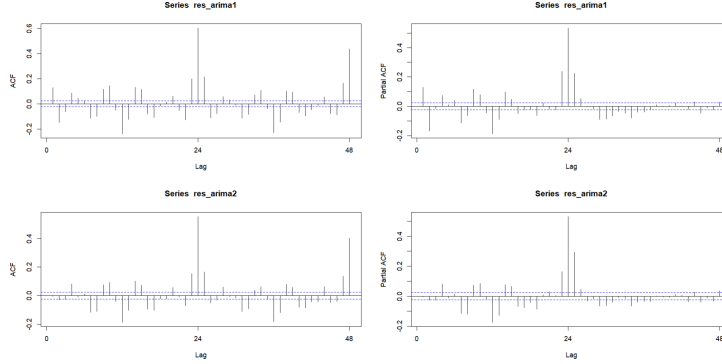


Figure 6: ACF and PACF of the residuals for the ARIMA(2,0,0) (above) and ARIMA(2,0,1) (below).

These results indicate that the dependence structure of the series is not adequately captured by non-seasonal ARIMA models alone, as we expected. Consequently, the analysis proceeds with Seasonal ARIMA (SARIMA) models, which explicitly account for the seasonal dependence observed in the data. In particular, the SARIMA(2,0,0)(1,0,0)[24] model is the first estimated as a baseline seasonal extension of the ARMA structure. To verify whether a moving-average component could improve the fit, a SARIMA(2,0,1)(1,0,0)[24] specification was also estimated. The estimated MA coefficient was found to be very close to zero and no substantial improvement was observed in the residual diagnostics. Both the residual series exhibit significant autocorrelation, with particular spikes around lag 24, the Ljung–Box test rejected the null hypothesis of white-noise residuals and the AIC value is nearly identical for the two specifications (129008.8 and 129008.7 respectively).

Finally, an automatic model selection procedure was performed using the function `auto.arima()`. The selected specification was SARIMA(2,0,0)(2,1,0)[24] and the AIC value of 128931 is significantly better than the previous two SARIMA models.

However, the residual diagnostics indicate that part of the temporal dependence remains unexplained. In particular, significant residual autocorrelation is still observed at several seasonal lags, including the same as for the previous models (around lag 24) and at lag 48. The Ljung–Box test rejects the null hypothesis of white-noise residuals, as confirmation to our correlograms (Figure 7).

Therefore, although the SARIMA models provide a better representation of the seasonal structure than the corresponding non-seasonal ARIMA specifications, they are not able to fully capture the dependence structure of the series. This suggests that additional information may be required to explain part of the remaining variability in hourly demand.

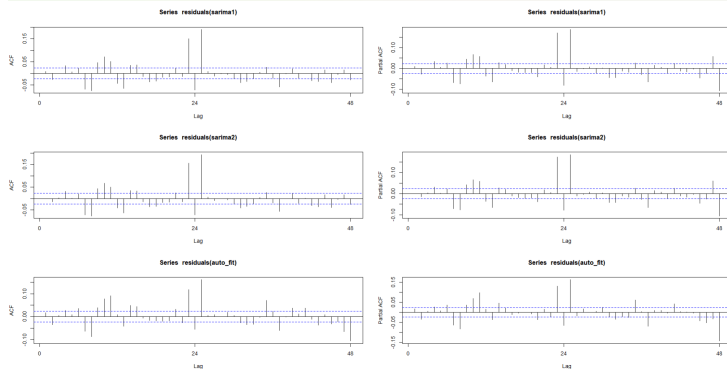


Figure 7: ACF and PACF of the residuals for the SARIMA(2,0,0)(1,0,0)[24] (above), SARIMA(2,0,1)(1,0,0) [24] (in the middle) and SARIMA(2,0,0)(2,1,0)[24] (below).

4.4 SARMAX

In this section, models with exogenous regressors are considered. Since the exploratory analysis has already highlighted a strong and persistent seasonal structure in the data, we focus directly on SARMAX specifications, which allow for the joint representation of both seasonal dynamics and external explanatory variables.

The inclusion of exogenous variables is consistent with the preliminary correlation and multicollinearity analysis, which indicated that the selected regressors do not exhibit problematic levels of redundancy and can therefore be included in the model specification. We defined the time series with a frequency of 24 and compared two SARMAX specifications: a manually selected SARIMA(2,0,0)(1,0,0)[24] model with exogenous regressors and an automatically selected model obtained with `auto.arima()`, resulting in a SARIMA(1,0,4)(2,1,0)[24] structure. The AIC value of the automatically selected model is 125580.1, with a better fit than the manually selected model that has an AIC value of 125837.8.

The inclusion of external covariates substantially improved the predictive performance relative to the SARIMA models of the previous section. Most explanatory variables were statistically significant and displayed economically meaningful signs. In particular, higher toll costs (**Avg_tolls**) and airport fees (**Avg_airport_fee**) were associated with lower demand, suggesting a negative relationship between trip costs and ride requests. Conversely, **Shared_trips** and **Avg_congestion_surcharge** exhibited positive coefficients, indicating that periods characterized by higher traffic intensity and shared mobility activity tend to coincide with stronger demand.

However, not all variables behaved consistently across specifications. In particular, **Avg_base_fare** was not statistically significant in either model ($t = -1.21$ in the manually specified SARMAX and $t = 1.58$ in the automatic specification). This suggests that the effect of the base fare is not clearly distinguishable once the temporal dynamics and the remaining regressors are taken into account.

Despite the explanatory improvement, the residual diagnostics reveal that important temporal dependence remains unexplained. For both specifications, the Ljung–Box test strongly rejected the null hypothesis of white-noise residuals. Furthermore, the residual ACF plots continued to display several significant spikes at seasonal lags (Figure 8), indicating that a substantial portion of the seasonal structure remains unmodeled.

Therefore, although the exogenous regressors explain an important share of the variation in demand and substantially improve predictive accuracy relative to the SARIMA models, they do not eliminate the residual autocorrelation. The residual patterns are remarkably similar to those observed for the SARIMA specifications, suggesting that the main limitation is not the absence of explanatory variables but rather the inability of a single-seasonality framework to fully represent the complex temporal dynamics of hourly demand.

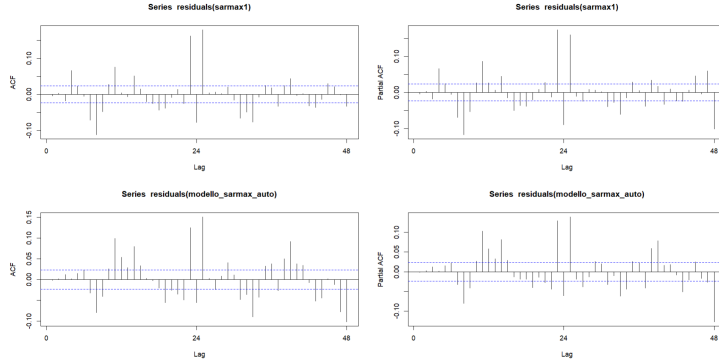


Figure 8: ACF and PACF of the residuals for the SARMAX(2,0,0)(1,0,0)[24] (above) and SARIMA(2,0,0)(2,1,0)[24] (below).

The information gathered from both SARIMA and SARMAX models suggests that the remaining dependence is primarily associated with seasonal effects that are not adequately captured by a single 24-hour cycle. Consequently, a more flexible approach is required to include the multiple temporal scales that characterize the hourly demand of taxis. For this reason, the next section introduces a Fourier-based specification which allows daily and weekly periodicities to be modeled simultaneously while maintaining the explanatory power of the external regressors.

4.5 Fourier Dual-Seasonality Model

To account for the presence of multiple seasonal cycles, we extend the previous specifications by introducing a dual-seasonality Fourier representation. In particular, we model both the daily (24-hour) and weekly (168-hour) periodic components through Fourier terms, which are included as exogenous regressors within an ARIMA framework.

The final specification is selected via `auto.arima()`, resulting in an ARIMAX(4,1,2) model with a set of Fourier regressors and economic covariates. This approach

allows for a flexible and smooth representation of seasonal patterns without imposing rigid seasonal differencing structures.

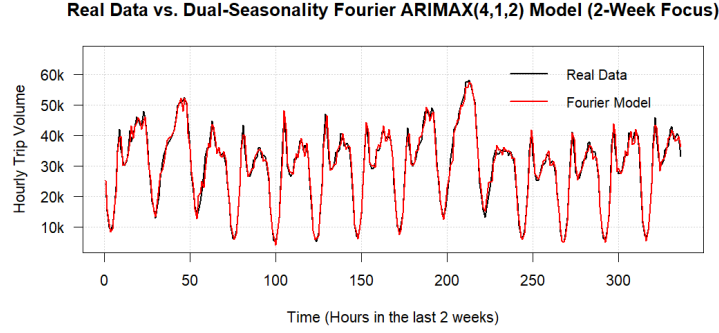


Figure 9: Real Data vs Fourier Model

The inclusion of economic regressors further improves interpretability, with coefficients generally consistent with economic intuition: tolls and airport-related fees exert a negative effect on demand, while shared trips and congestion-related surcharges are positively associated with higher demand levels.

However, the residual diagnostics indicate that the dependence structure is not fully eliminated. While short-term autocorrelation is effectively removed (with $ACF_1 \approx -0.001$), the residual ACF still exhibits significant spikes, particularly at lag 168 and 336. This suggests that, although the Fourier representation captures a large portion of the seasonal dynamics, part of the weekly structure remains only partially explained.

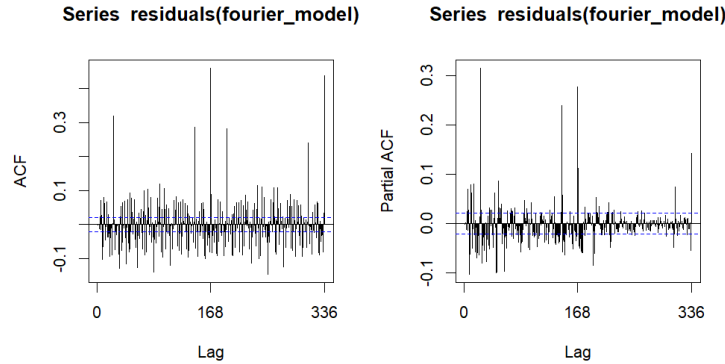


Figure 10: ACF and PACF for The Fourier Model

Overall, the Fourier specification represents an improvement over previous models, but it still does not fully account for the complex temporal dependence

present in hourly taxi demand.

4.6 Generalized Additive Model (GAM)

To further investigate the behavior of the factors influencing taxi demand, we looked for potential non-linear relationships between the regressors and the dependent variable. Preliminary linearity checks revealed that several predictors did not have a linear relationship with the response. Because of this, a Generalized Additive Model (GAM) was specified, which allows for both linear and non linear effects of the regressors to be modelled.

Following this data-driven approach, variables were selectively incorporated into the model according to their specific structural and statistical properties. Continuous economic factors, specifically average trip time, tips, airport fees, and the volume of accessible trips (`wav_trips`), were modeled using non-parametric smoothing splines. Conversely, specific predictors were intentionally restricted to a parametric linear form to maintain model parsimony and avoid overfitting. The average tolls variable was kept linear, revealing a highly significant negative impact on demand, while the congestion surcharge and shared trips were also included linearly. The positive coefficient of the congestion surcharge captures a clear simultaneity effect, as this fee naturally triggers during peak traffic hours when passenger volume is at its highest. Furthermore, the hour of the day was modeled using a cyclic smoothing spline to mathematically guarantee a smooth, continuous transition between the end of one day and the beginning of the next, while the weekday variable was integrated as a random effect.

The GAM model delivers an adjusted R^2 of 0.275, explaining 27.6% of the total deviance. While the model successfully captures the structural temporal effects and several key linear regressors, other non-linear splines for economic factors do not reach statistical significance, suggesting that these specific drivers may influence demand in ways not fully captured by this additive structure or that they interact with temporal dynamics in more complex ways. Despite its significance, residual diagnostics still display a structured pattern (Figure 11). The Ljung-Box test on the GAM errors strongly rejects the null hypothesis of random noise. This proves that while the GAM successfully extracted the intricate price-sensitivity dynamics and structural patterns, a substantial amount of short-term temporal dependence and stochastic memory remains trapped within the errors.

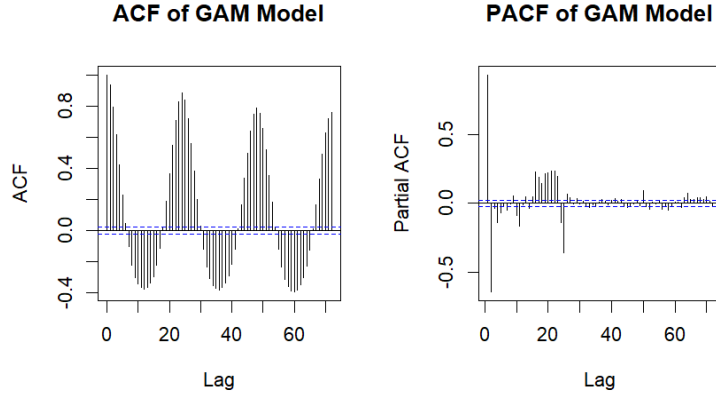


Figure 11: ACF and PACF of GAM Model

Consequently, this remaining autocorrelation justifies the adoption of a hybrid GAM-SARIMA architecture. By fitting a SARIMA(2, 0, 0)(1, 0, 0)[24] model directly to the GAM residuals, we aim to capture these residual temporal dynamics and systematically filter the error term. This step is designed to extract the dataset's remaining short-term and daily cyclical memory, pushing the final residuals closer to a true white-noise process prior to evaluating the model's actual forecasting performance on the out-of-sample test set.

It is worth noting that, given the large number of observations available, the Ljung-Box test is highly sensitive and may reject the null hypothesis of white noise even when the remaining autocorrelation is practically negligible from a forecasting perspective. For this reason, model adequacy is assessed not only through formal hypothesis testing, but also through residual plots and forecasting accuracy measures.

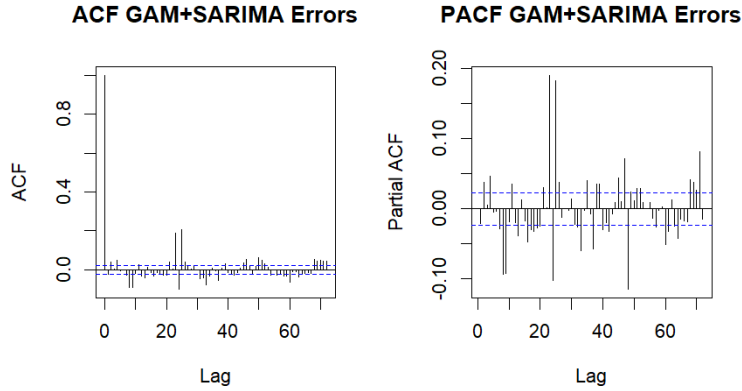


Figure 12: ACF and PACF of GAM + SARIMA errors

5 Conclusion

The comprehensive evaluation of the forecasting models used in our project is presented in the table below, highlighting the necessity of considering both exogenous variables and seasonality for the adequate prediction of taxi rides demand. The models were compared based on their Root Mean Square Error (RMSE) evaluation metric on both the training and test sets, considering also their Gap RMSE to monitor the presence of overfitting.

Model	RMSE train	RMSE test	Gap RMSE
ARIMA(2,0,0)	3083.76	12013.84	8930.08
ARIMA(2,0,1)	3021.77	12013.03	8991.26
SARIMA(2,0,0)(1,0,0)[24]	2400.87	11859.60	9458.73
SARIMA(2,0,1)(1,0,0)[24]	2400.52	11860.57	9460.05
SARIMA(2,0,0)(2,1,0)[24]	2461.30	8320.91	5859.61
SARMAX(2,0,0)(1,0,0)[24]	1912.62	9062.13	7149.51
SARMAX(5,0,1)(2,1,0)[24]	1933.33	5387.35	3454.02
Fourier-ARIMAX(4,1,4)[24, 168]	1784.61	4632.40	2847.79
GAM (Structural)	9572.18	11024.10	1451.92
GAM + SARIMA(2,0,0)(1,0,0)[24]	2052.85	10863.25	8810.40

Table 1: Performance metrics for different models using the Root Mean Square Error (RMSE)

Evaluating the results, the limitations in the usage of pure autoregressive models become evident. It is worth noticing that the ARIMA models, both ARIMA(2,0,0) and ARIMA(2,0,1), as well as the SARIMA models, have shown the worst results on the test set, with the RMSE values consistently above 11800, a part for the SARIMA(2,0,0)(2,1,0)[24], whose value still remains high. At the same time, they are subject to overfitting, as evidenced by a gap RMSE above 8900. Therefore to capture the taxi demand dynamics is not sufficient relying on past lags and simple seasonal terms.

The addition of external economic regressors marks a clear turning point in improving predictive accuracy. For instance, the SARMAX(5,0,1)(2,1,0)[24] model drastically reduces the RMSE on the test set to 5387.35. Providing the algorithm with some information regarding real-world phenomena such as trip times, fares, and tolls, increases the capacity of generalization substantially, thus allowing one to avoid overfitting by 3454.02.

The analysis on the Generalized Additive Model (GAM) provides additional insights. For instance, when looking at the pure GAM without modeling the residuals, it is possible to see that its gap is only 1451.92, which means stability; however, it is underfitting since it has a high test error of 11024.10. When trying to model the residuals using SARIMA errors, it becomes possible to minimize the test error to 10863.25; however, the Gap RMSE goes up again to 8810.40. This dynamic suggests that the stochastic noise captured in the training residuals does not hold consistent, repeatable patterns in the future test horizon.

Ultimately, the Fourier ARIMAX becomes an best-performing model in terms of accuracy of performance. The combination of its ability to capture the presence of complicated seasonality patterns in time series data using the Fourier approach along with its capability to include the influence of exogenous variables gives it the smallest possible test RMSE value of 4632.40. Additionally, it demonstrates the best generalization abilities among high-performing models with an RMSE gap of 2847.79. Overall, this model provides an excellent balance between mathematical flexibility and predictive stability, successfully decoding both the daily and weekly cycles as well as external economic shocks to deliver highly reliable and consistent demand forecasts.

Limitations and Future Work

While our study found effective models for taxi demand forecasting, a few limitations should be mentioned. First, we analyzed New York City as a single, large macro-area. Splitting the demand by specific neighborhoods or taxi zones would certainly help to capture local mobility patterns more accurately. However, we kept a city-wide approach based on the available data and our primary goal of predicting the overall aggregate demand.

Secondly, looking at the model diagnostics, we did not achieve perfect white-noise residuals, even with our best hybrid models. However, this is expected when dealing with hourly urban mobility data. Real-world taxi demand is constantly affected by unpredictable local events, such as sudden weather changes, traffic jams, or spontaneous social gatherings. Broad economic variables and standard seasonal cycles simply cannot predict these random shocks. As a result, a certain amount of noise will naturally remain in the model’s errors.

Finally, future developments could explore the application of advanced Machine Learning and Deep Learning architectures, such as Extreme Gradient Boosting (XGBoost) or Long Short-Term Memory (LSTM) neural networks. These highly data-driven algorithms could potentially capture complex, multi-dimensional interactions between temporal cycles, economic shocks, and the remaining residual dynamics.